

Health & Social Care in Hansard: a topic-pilot brief

The pipeline, the pilot test, and the substantive checks we now need

July 2026

1. The goal

We want to measure what Parliament has said about health and social care (H&SC) across the full Hansard record, 1803 to the present. For every speech we want to know two things: whether it substantively engages H&SC, and if so which aspects of it the speaker raises. With that in place we can trace how the volume and the mix of H&SC debate shift over two centuries, by era and by topic. The obstacle is scale. The record runs to millions of speeches, far more than anyone can read, so the measurement has to be done by machine and then trusted. This note describes the pipeline we have built to do that, the pilot we ran to test it, and the substantive questions we now need domain input on before scaling up.

2. The main pipeline, step by step

The production pipeline has four stages. Each exists for a reason worth stating, because the reasons are also where the method could go wrong.

1. Narrow the corpus to plausible candidates. The large majority of Hansard never touches H&SC, so running a language model on every speech would be wasteful. In production this stage will be a semantic search over sentence embeddings, retrieving speeches close in meaning to H&SC whatever their exact wording, and that retrieval step will itself need validation. For now we use a broad keyword net (NHS and public-health terms for the modern era; Poor Law, workhouse, sanitary-reform and epidemic-disease terms for the nineteenth century) as a deliberate placeholder: it is crude, but it lets us build and test the rest of the pipeline without waiting on the retrieval work. Either way this stage decides only what gets looked at, never the answer, and it is deliberately over-inclusive. It leaves a few hundred thousand candidate speeches rather than millions.
2. Read each candidate with a language model. For every surviving speech the model answers three questions: does this speech substantively discuss H&SC (yes or no); if so, which sub-topics does it raise; and a supporting verbatim quote. The judgement needs to weigh meaning in context, which a keyword cannot do. The sub-topics are free text, written by the model rather than chosen from a fixed list, because the aim is to discover what is actually discussed rather than to score speeches against categories we set in advance.
3. Aggregate the free-text sub-topics into a taxonomy. The models describe the same idea in many wordings, so to count topics we group same-meaning phrases automatically and label each group. Section 5 explains how, and why this step is more delicate than it looks.
4. Turn the labelled topics into measures over time. The payoff is prevalence: how much of H&SC debate each topic occupies, and how that changes across the 220 years.

Stage 2 sends the model a short instruction, an output-format request, and the speech. The core elements are reproduced verbatim below; the pilot varies their surface form (Section 3) but

not their meaning. The topic definition is carried inside the instruction, which makes it easy to overlook as a piece of wording. It is in fact the most consequential setting in the pipeline, and Section 5 shows what happens when it changes.

Instruction (one of two paraphrases):

Determine whether the parliamentary speech below substantively discusses health and social care — that is, UK health and social care policy: the NHS, public and mental health, adult social care, care homes, and the people who provide or rely on that care — rather than merely mentioning it in passing. If it does, identify the specific sub-topics of health and social care that it discusses. Describe each sub-topic in your own words as a short phrase (a few words). Give at most 5 sub-topics. Also provide the single most relevant verbatim quotation from the speech as supporting evidence.

Output format (the structured condition):

Respond with a single JSON object and nothing else, with exactly these keys: “mentions_topic” (true or false), “subthemes” (an array of short free-text phrases, empty if mentions_topic is false), and “evidence_quote” (a string, empty if mentions_topic is false).

The other output condition gives no format instruction at all, leaving the model to answer in free text. When present, an optional system message adds a role: “You are an expert analyst of health and social care policy in the United Kingdom, with a deep understanding of parliamentary debate.”

3. The pilot: a test before scaling

Running stages 1 to 4 on the whole corpus is expensive and hard to undo, so we first tested the approach on a sample of 270 speeches, stratified across eras and speech lengths. The pilot adds one thing the production run will not use: a robustness harness. Instead of reading each speech once, we read it many ways and check that the answer is driven by the speech rather than by incidental choices of model or wording. If the answers move a lot under variation, the production numbers are not yet trustworthy.

Concretely, each speech was read by four models (three production candidates plus one larger reference model) under eight wordings of the same request, varying the role framing, the paraphrase, and the output format. That is 32 independent reads per speech. Two results frame everything that follows:

- On the yes/no presence question the reads agree with each other about 82% of the time, which is solid but short of unanimous.
- The share of speeches counted as being about H&SC ranges from 34% to 81% depending on prompt and model, averaging around 51%. Much of that swing comes from the output format: asking for free-text output puts presence at about 55%, while asking for structured output puts it at about 43%. The headline rate is therefore sensitive to prompt design, which is one reason the substantive checks below matter.

4. What the models found: the topic map

The topic map below is built from the configuration we now expect to ship: the expert healthcare-then-social-care definition, under structured (JSON) output, no role framing, and no five-topic cap. Across the 270 speeches that arm produced 1,412 distinct sub-topic phrases (1,673 emissions). Grouped by meaning, the most common topics are shown below. Both modern NHS-era themes and nineteenth-century themes such as Poor Law, sanitation and housing appear. This coverage should be read with the sampling in mind: the keyword net was deliberately built to include both

modern and historical H&SC vocabulary, so the presence of both is partly a consequence of how the sample was seeded rather than independent evidence about the balance of the corpus. What the map does show is what the models surface as sub-topics once a speech is placed in front of them, and that is what the rest of this note examines.

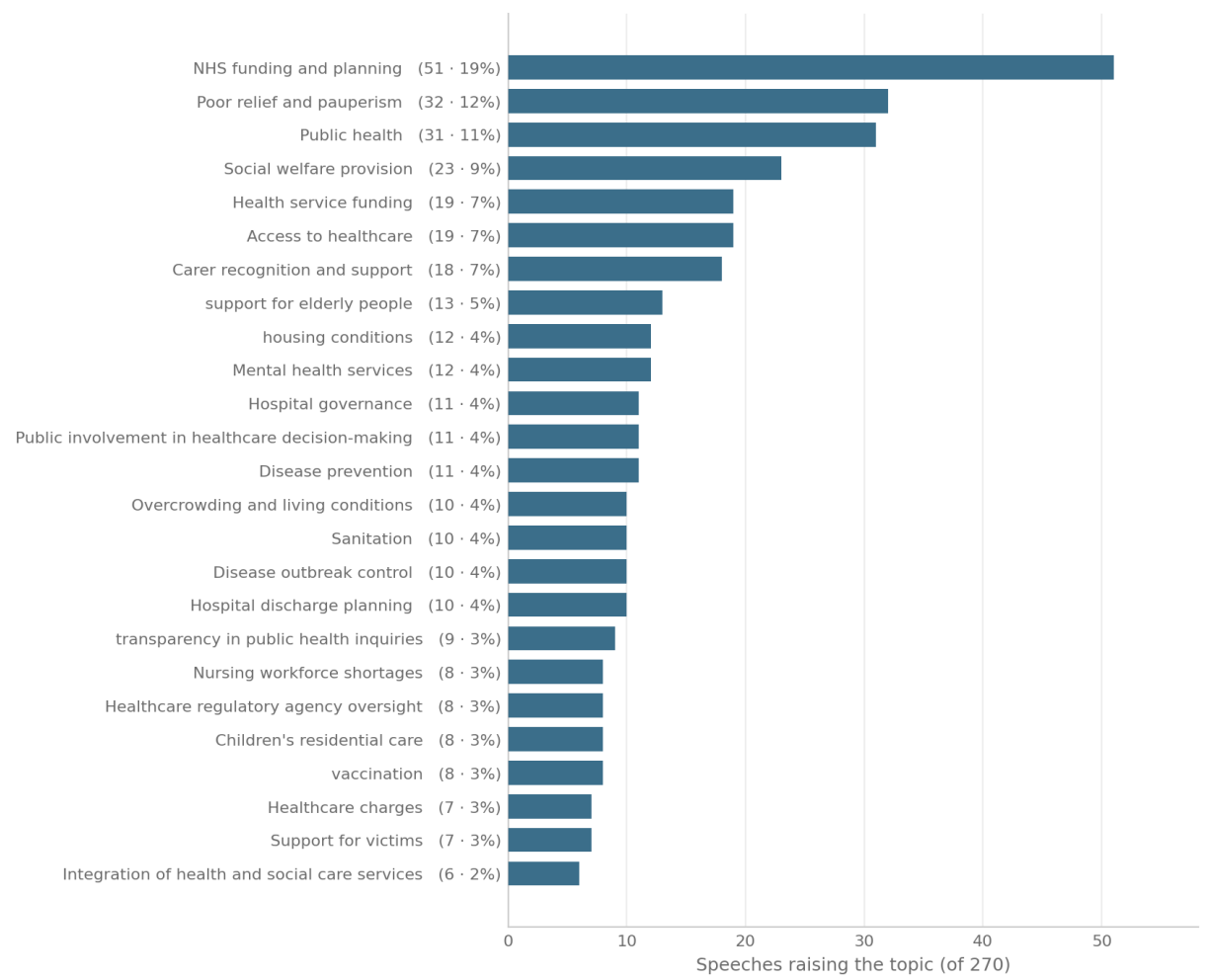


Figure 1: Most common H&SC topics under the expert HC→SC definition with structured output (share of speeches raising each; count in brackets). Labels are the clusters’ automatically chosen medoid phrases, not hand-edited.

Each row below is a machine-discovered topic, that is, a cluster of same-meaning phrases, ranked by how many of the 270 speeches raise it. The “Phr.” column counts how many distinct wordings the models used for that one idea. The label in each row is the cluster’s medoid: the single emitted phrase closest to the cluster’s centre in meaning-space, chosen automatically rather than written by us. Nothing here is hand-edited, which is why some labels read awkwardly or are oddly cased (for example “support for elderly people” or “housing conditions”). Naming the topics well is one of the domain judgements we are asking for.

#	Merged topic	Speeches	Phr.	Example sub-themes the models wrote
1	NHS funding and planning	51 (19%)	112	NHS funding/performance; NHS resource management; NHS funding
2	Poor relief and pauperism	32 (12%)	79	Poor Law relief; Poor law/pauperism; Poverty and the Poor Law
3	Public health	31 (11%)	41	Public Health Policy; Public Health Measures; Public health infrastructure

#	Merged topic	Speeches	Phr.	Example sub-themes the models wrote
4	Social welfare provision	23 (9%)	33	Social welfare support; Social services funding; Social welfare governance
5	Health service funding	19 (7%)	29	Healthcare funding; public health funding; public financing of health services
6	Access to healthcare	19 (7%)	27	Access to healthcare services; Care and treatment access; Healthcare access
7	Carer recognition and support	18 (7%)	46	Social care funding; Care for carers; Public funding for care
8	support for elderly people	13 (5%)	37	Care for pensioners; Old age support; Elderly care
9	housing conditions	12 (4%)	30	Social housing; Public health housing; Housing standards
10	Mental health services	12 (4%)	29	Hospital provision for mental health; Mental health provision; mental health facilities
11	Hospital governance	11 (4%)	24	Hospital management and administration; Hospital administration and efficiency; hospital management
12	Public involvement in healthcare decision-making	11 (4%)	26	Decentralisation of healthcare decisions; Public involvement in healthcare; Patient involvement in healthcare
13	Disease prevention	11 (4%)	12	preventive healthcare; Preventative healthcare; Preventive Care
14	Overcrowding and living conditions	10 (4%)	15	Overcrowded housing; Barrack conditions; Poor living conditions
15	Sanitation	10 (4%)	19	Sanitary conditions; Sanitary conditions in public buildings; Sewage and sanitation
16	Disease outbreak control	10 (4%)	24	Epidemic control; Epidemic response; Infectious disease response
17	Hospital discharge planning	10 (4%)	17	hospital care; Hospital discharge; Hospital preservation
18	transparency in public health inquiries	9 (3%)	14	Transparency of public health bodies; Public trust in healthcare delivery; transparency in health decision-making
19	Nursing workforce shortages	8 (3%)	19	Workforce crisis in healthcare; Workforce crisis in health; Nursing workforce
20	Healthcare regulatory agency oversight	8 (3%)	14	Healthcare agencies; Health commission; Department of Health oversight
21	Children’s residential care	8 (3%)	19	Children in care; child protection services; protection of young children in care settings
22	vaccination	8 (3%)	22	vaccination and immunisation; immunisation; vaccination policy
23	Healthcare charges	7 (3%)	19	Private fees for medical practitioners in pay beds; Patient charges/costs; Pay beds and private fees in public hospitals
24	Support for victims	7 (3%)	8	Support for the vulnerable; support for recovery; Support for vulnerable immigrants
25	Integration of health and social care services	6 (2%)	20	Patient-centred services; Integrated care; Patient-centred care services
26	Food poverty	6 (2%)	7	Food costs and access; Food supply shortages; Food poverty and food bank support
27	Social worker training and support	6 (2%)	13	Care worker training; Social worker training; Health professional training and support
28	tuberculosis treatment	6 (2%)	13	TB care and treatment; Tuberculosis control; TB treatment
29	Medical aid provision	6 (2%)	10	Medical aid; Healthcare supply chain; Medical Relief
30	Independence of the medical profession	6 (2%)	16	Doctor compensation and retirement; Medical profession independence; Doctor recruitment to under-doctored areas
31	Well-being as a guiding principle	6 (2%)	7	Well-being as guiding principle; Mental and physical well-being in care; Physical well-being and basic living standards
32	Government funding allocation	6 (2%)	6	Public funding allocation; Funding and expenditure oversight; funding and budget allocation
33	Parliamentary scrutiny of care regulations	5 (2%)	7	Reform of healthcare and social care regulatory bodies; parliamentary scrutiny of health regulations; Timing of health and social care legislation

#	Merged topic	Speeches	Phr.	Example sub-themes the models wrote
34	Learning disability nursing	5 (2%)	15	nursing care; Learning disability nursing shortages; Mental health nursing
35	Health and working conditions	5 (2%)	11	Health of working life; Workplace health risks; health impacts of poor working conditions
36	Mental capacity legislation	5 (2%)	11	Mental health and capacity law; Mental Capacity; Mental capacity and decision-making
37	Health visitor roles	5 (2%)	9	Health visitor role; Health visitor public health role; Service personnel healthcare access
38	disease diagnosis	5 (2%)	8	disease detection and diagnosis; Diagnosis of illness; health diagnostics
39	Taxation on low-income households	5 (2%)	8	Taxation and its impact on low-income groups; Local taxation for relief; Local fiscal burden and taxation
40	Access to social support	5 (2%)	4	Social integration; Social needs prioritization; Social clubs

A long tail of smaller topics sits below this list and grows increasingly specific: 152 clusters in total, of which 50 are raised by four or more speeches. Ranks 41 onward are in the appendix.

5. Where the sub-topics come from, and why aggregation is delicate

The topic map is not a neutral readout of what Parliament said. It is shaped at several points by choices we made, each of which the pilot suggests we should revisit with domain input.

The definition sets the era profile. The presence and era findings below compare several construct wordings; the topic map in Section 4 uses the expert healthcare-then-social-care wording we expect to ship. The original pilot definition, quoted in Section 2, names the NHS, adult social care and care homes. Those are institutions that did not exist for the first century and a half of the record, so the wording risks reading the nineteenth century as quiet about health simply because it lacked the vocabulary. We tested this by re-running the same speeches, models and formats against an era-neutral rewrite that describes the function rather than the institution: the health of the population and the care of people who are sick, injured, disabled, elderly or destitute, however that care is provided and paid for. The effect is large, and concentrated exactly where the concern predicted. Under the current definition the share of speeches judged H&SC climbs from 33% before 1900 to 62% after 1948; under the era-neutral wording it runs from 54% to 67%, and the rise across the record falls from 29 to 13 percentage points (Figure 2). The two definitions agree on 82% of individual reads, and the disagreements run almost entirely one way: the era-neutral wording adds speeches, in 23% of pre-1900 reads but only 7% of post-1948 ones. On this evidence a good part of the apparent growth in health debate after the NHS is an artefact of how we asked the question.

This does not establish that the era-neutral wording is the right one. Reading the speeches it newly admits shows both kinds of case. Some are plain misses by the current definition: an 1868 exchange in which the Poor Law Board sets standards for the diet, lodging and work of vagrants is unmistakably the Victorian care system, and the current wording scored it as H&SC in only one read out of eight. Others look like over-reach: an 1856 debate about which body should hold powers under an agricultural statistics bill was counted as H&SC by seven of eight reads because the Poor Law Board happens to be named in it, which is precisely the passing mention the instruction rules out. That second case is worth stating plainly, because it is the same failure as before rather than a new one. The old wording triggered on the token NHS; the new one triggers on tokens like Poor Law Board. Changing the definition moved which century the anchoring flatters, but did not stop the models keying on institution names instead of on whether care is the subject of the speech. Choosing between the two wordings, and repairing whichever we keep so that neither trap fires, is a domain judgement and not something a further run can settle.

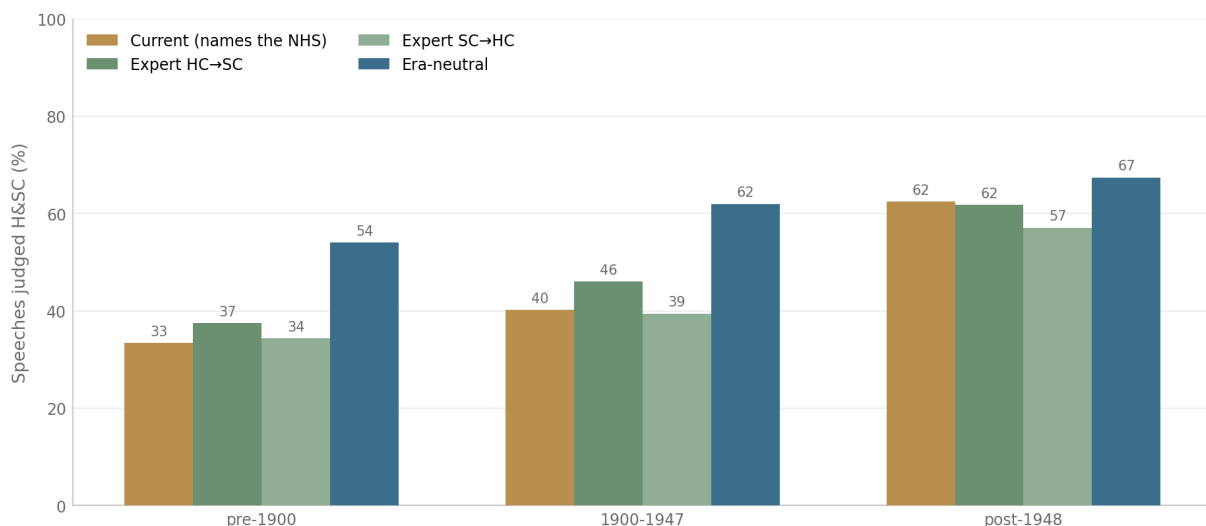


Figure 2: Share of speeches judged to be about H&SC, by era, under four construct definitions. Neither is known to be correct; the point is that the choice of wording, not the record, sets much of the slope.

An expert definition of both healthcare and social care lands close to current, not to era-neutral. Domain experts supplied separate sentences for healthcare (“primarily concerned with the prevention, diagnosis and treatment of illness”) and social care (“practical support to people who need assistance with everyday living...”). Because the construct of interest is speeches about *both*, the two sentences were concatenated and run as a new definition arm under the same shipping default (no role, no cap, both formats, all models). Two orders were tested: healthcare then social care, and social care then healthcare. The headline rates sit next to current rather than next to era-neutral: 51% under HC→SC and 46% under SC→HC, against 48% for current and 62% for era-neutral. Matched-cell agreement with current is 88% and 87% respectively; speech-level majority labels agree with current on 96% and 93% of speeches. The era profile tells the same story (Figure 2): both expert orders keep a steep post-1948 rise (+24 and +23 percentage points from pre-1900), close to current (+29) and far from era-neutral (+13). So replacing the pilot wording with the expert sentences does not, by itself, undo the NHS-era climb that the earlier definition contrast flagged as partly artefactual. The order of the two sentences is a real but secondary effect: HC-first is about five points more inclusive than SC-first on matched cells (89% agreement; 166 cells yes only under HC→SC versus 62 only under SC→HC), and only 17 of 270 speeches flip their majority label. Free-text output is more order-sensitive than structured output; one model (Llama-70B) is almost order-invariant. If a single expert wording is shipped, HC→SC is the closer twin of the current pilot definition. The contested speeches under each pair — expert-order flips, and current versus either expert arm — can be inspected in `definition_review.html`, which now lets the reader switch comparison presets.

The expert definition changes who is counted more than what is named. Sub-topics were collected on every positive read of the definition arm, so the maps can be compared directly. Holding format fixed at structured output (the cleaner stream, and the one we intend to ship), the expert HC→SC map and the current-definition map agree on twenty of the top twenty-five topics by meaning (embedding match), and the ranks of those matched topics correlate at Spearman 0.83. The headline families stay put: NHS funding, public health, Poor Law / pauper relief, social welfare, access to care, mental health, housing and sanitation. What moves is emphasis at the margin. Under the expert wording, carers and elderly support rise into the top ten, and disease prevention, overcrowding and healthcare-regulator oversight enter the top twenty-five; under the current wording, local governance of health services and health infrastructure sit higher. Absolute

speech counts are a little lower than in the earlier full-grid map because this comparison uses one definition $\times$ one format rather than eight prompt wordings, but the shape of the taxonomy is stable enough that swapping the shipping definition does not force a fresh substantive review of the topic list from scratch — only a check on the handful of families that shift rank.

The prompt’s output format changes the result. Each read was asked either for free-text output or for structured output, and this matters more than expected. The number of sub-topics per speech is unaffected (about 3.8 either way, often hitting the cap of five), so the format does not squeeze the count. What it changes is the wording and the presence rate. Free-text output produced 5,903 distinct phrasings; structured output produced only 3,639, roughly 40% fewer, because structured output regularises how the model names the same idea while free text lets it drift. Free-text output also raised the presence rate (55% versus 43%) and failed to parse cleanly about 5% of the time, which is the source of the noise described below. So the raw phrase counts in the table, and how badly topics fragment, are partly a property of prompt design rather than of the speeches. In production we will fix a single output format, and that is itself a measurement decision, not just an engineering one.

The five-topic cap inflates the count. The instruction asks for at most five sub-topics, and we have now tested removing it: the same speeches, models and formats, run once with the “at most five” sentence and once without. The cap turns out to work as an anchor, not just a ceiling. With the cap, half of all positive reads return exactly five sub-topics and the median is five. Without it, that spike disappears (only 9% land on five), the distribution settles onto a smooth curve centred on two to three, and the median drops to three (Figure 3). About 14% of speeches genuinely warrant more than five sub-topics when allowed, up to eighteen, but those are the minority; the dominant effect is the cap pulling the many lower-content speeches up towards five. Presence is unaffected (48% either way), so this is purely about how many sub-topics get counted, not whether a speech is judged to be about H&SC. The practical implication is that any count-based measure built on the capped runs is biased upward, so production should drop the cap or set it far higher.

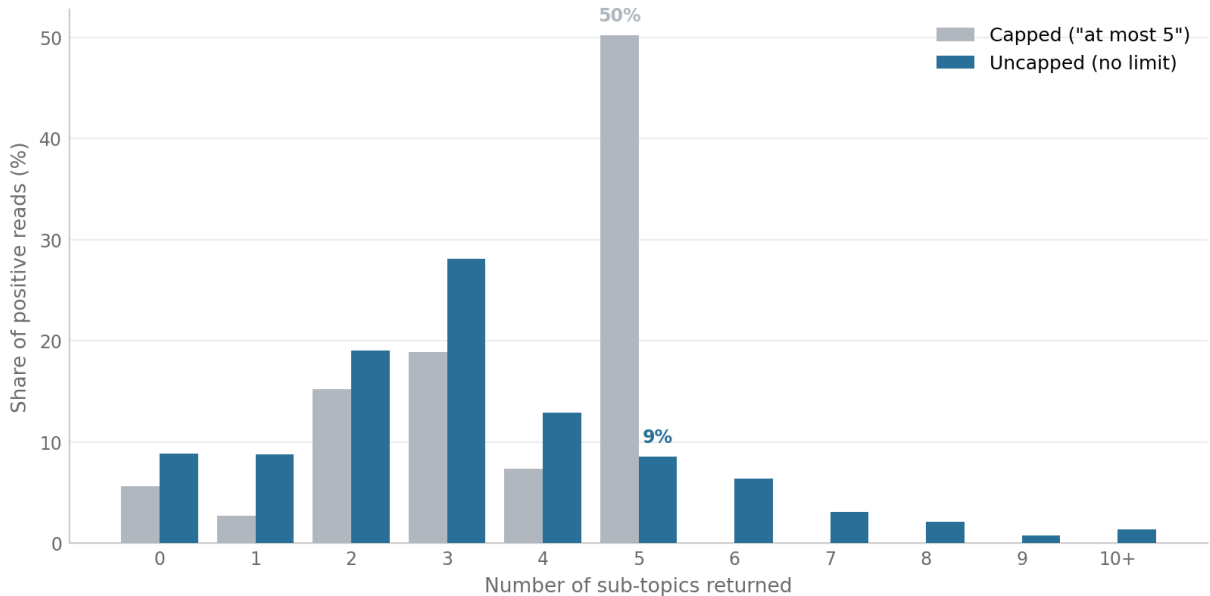


Figure 3: Sub-topics returned per positive read, capped versus uncapped, on the same speeches. The cap concentrates half of all reads at exactly five; without it the count follows a smooth low distribution.

Aggregation depends on one threshold. Same-meaning phrases are merged by converting

each phrase into a numeric representation of its meaning (an embedding) and clustering the ones that sit close together, with each cluster labelled by its most central phrase (its medoid). A single setting controls how readily phrases merge. Set it loose and unrelated ideas get lumped together; set it tight and one idea fragments into several topics. We currently keep it tight to avoid false merges, and the cost of that is visible in the table: NHS funding and planning (row 1) sits apart from Health service funding (5) and Carer recognition and support / social care funding (7); Public health (3) sits apart from Disease prevention (13) and Disease outbreak control (16); and Hospital governance (11), transparency in public health inquiries (18) and Healthcare regulatory agency oversight (20) split what could be one governance family. Where these should be merged, and how broad the final families should be, is a domain judgement rather than a distance measurement.

A small share of entries are formatting noise, not content. From the free-text parse failures above, a model’s answer sometimes leaked into the topic field. These are a known and fixable side effect, not real findings, but they are worth seeing:

- Does the speech substantively discuss health and social care? No
- Sub-topics of health and social care discussed: None
- Most relevant verbatim quotation (for context): ...

Taken together, the prevalence numbers depend jointly on the definition, the output format, the five-topic cap, and the clustering threshold. We can now put rough sizes on the first two: switching from the current wording to era-neutral moves the era profile by around 16 percentage points of slope, while swapping in the expert healthcare-plus-social-care wording leaves that slope largely intact (+23 to +24 pp rather than +29) and moves the headline rate by only a few points; presentation order within the expert wording adds about another 5 points. The output format moves the headline rate by about 12 points under every definition tested. None of these is settled by the machine, which is precisely why the next step is substantive rather than technical.

6. What we need next

Two pieces of work now need domain input, and they are the reason for this brief.

The first is a substantive review of the current map:

1. Scope. Section 5 shows the definition is the most consequential setting we have found, and the pilot cannot resolve it alone: the era-neutral wording fails in the opposite direction from the current one, while the expert healthcare-plus-social-care wording tracks current closely and does not remove the post-1948 climb. Which bound is closer to how the field thinks of H&SC, should the expert sentences be preferred as the shipping text (and in which order), and how would you word the construct so that neither the NHS nor the Poor Law Board pulls a speech in on the strength of being mentioned?
2. Completeness. Is any important H&SC theme missing that Parliament would have debated across this period?
3. Correctness. Is any listed topic wrong, or wrongly counted as H&SC?
4. Grouping and grain. How far should the over-split topics in Section 5 be merged, and at what level of detail should the final set sit?

The second is designing how the production pipeline will be evaluated, which the paper will stand or fall on. The pilot only shows that the models agree with each other, which is consistency, not that they agree with the field, which is validity. To close that gap we need to build, together:

- A gold standard: a set of speeches hand-labelled by domain experts for presence and for sub-topics. This is the reference the whole system is measured against, and the definition experiment has already produced an efficient starting set. On 159 of the 270 pilot speeches the two definitions reach different verdicts, 92 of them by a wide margin, and those are the speeches where the boundary is actually being decided rather than being obvious. They

are laid out in `definition_review.html`, an offline page that shows each speech next to what the models said under both definitions and records a yes, borderline or no judgement with a note, then exports the lot as a spreadsheet. The same page now also covers the expert-definition arm: a preset switches the columns to HC→SC versus SC→HC (or either expert arm against current), so the 17 majority-label order flips and the larger current-versus-expert disagreements can be reviewed the same way. Working through the contested set is worth considerably more per speech than labelling a random sample, in which most cases are easy and carry little information.

- An evaluation protocol we can re-run unchanged whenever a model, prompt, or threshold changes: presence scored against the gold labels, and sub-topics scored on whether the model’s topics match the experts’ in meaning.
- A decision, informed by that protocol, on the production settings the pilot has shown to matter: the definition, the output format, the sub-topic cap (Section 5 shows the current five is biasing counts), and the clustering grain.

Any form works for a first pass: notes in the margin, a shared document, or a call.

Appendix: reference

A. Filter keywords. The placeholder net from Section 2 matches the terms below, as word stems, in three groups. A speech enters the sample if it matches any of them; the terms decide only what is looked at, never the answer.

- Modern (NHS and welfare state): social care, care home, domiciliary care, residential care, adult social care, care worker, carer, NHS, national health service, public health, mental health, health care, healthcare.
- Pre-NHS (Poor Law and sanitary reform): poor law, workhouse, pauper, board of guardians, relieving officer, district nursing, lunatic asylum, feeble-minded, mental deficiency, friendly society, infirmary, almshouse, sanitary, board of health, fever hospital, vaccination, small-pox, cholera, tuberculosis, medical officer of health, dispensary.
- Epidemic and pandemic: covid, coronavirus, pandemic, long covid, lockdown, furlough, epidemic, influenza, spanish flu, quarantine, outbreak, typhoid, typhus, diphtheria, scarlet fever, plague.

B. Prompt, second paraphrase. Section 2 gives the first wording of the instruction. The pilot also uses this second, meaning-preserving paraphrase:

Read the parliamentary speech below. Decide if health and social care is a substantive subject of the speech and not just a passing reference. Treat health and social care as: UK health and social care policy: the NHS, public and mental health, adult social care, care homes, and the people who provide or rely on that care. When it is a substantive subject, list the particular aspects of health and social care it covers, each as a brief free-text label of a few words. Provide no more than 5 such labels. Include a short supporting quotation taken word-for-word from the speech.

Role and output format vary as in Section 2 (role present or absent, output structured or free text), giving eight prompt combinations in all.

C. Definition wordings tested. Each is dropped into the same slot in the instruction, after “substantively discusses health and social care, that is,”. Everything else in the prompt is identical, so the contrast between them is the definition and nothing else.

Current: UK health and social care policy: the NHS, public and mental health, adult social care, care homes, and the people who provide or rely on that care.

Era-neutral: the health of the population and the care of people who are sick, injured,

disabled, elderly or destitute, however that care is provided and paid for, whether by the state, local authorities, hospitals, charities, religious bodies or families.

Expert healthcare: Healthcare is primarily concerned with the prevention, diagnosis and treatment of illness.

Expert social care: Social care provides practical support to people who need assistance with everyday living because of age, disability, chronic illness or other long-term needs. Typical services include residential care, care homes, home care, support for carers and services for adults with disabilities.

The expert arm concatenates those two sentences in both orders (HC→SC and SC→HC) so that order effects can be measured without inventing new wording. Two further wordings prepared earlier, one narrower (clinical services only) and one broader (including the social determinants of health such as housing, sanitation and nutrition), are still available in config but have not been run; they would bracket the construct so the headline prevalence can be reported as a range rather than a single number.

D. Extended topic list, ranks 41 onward. This continues the table in Section 4. Of the 152 machine-discovered topics under the expert HC→SC / JSON arm, 50 are raised by four or more speeches.

#	Merged topic	Speeches	Phr.
41	public health guidance	5 (2%)	7
42	Healthcare policy	4 (1%)	4
43	impact of environmental hazards on health	4 (1%)	8
44	Patient safety	4 (1%)	5
45	NHS medical job opportunities	4 (1%)	9
46	hospital infrastructure development	4 (1%)	16
47	Medical education funding	4 (1%)	7
48	Medication compliance	4 (1%)	13
49	Basic salary for doctors	4 (1%)	10
50	Professional accountability in healthcare	4 (1%)	8
51	Nurse regulation	3 (1%)	13
52	Disability provision	3 (1%)	7
53	Maternity care	3 (1%)	3
54	suicidal ideation	3 (1%)	6
55	National health standards	3 (1%)	6
56	emergency care demand	3 (1%)	6
57	National Insurance contributions	3 (1%)	4
58	Access to food services	3 (1%)	4
59	Racial inequality in healthcare	3 (1%)	5
60	Service Shaping	3 (1%)	3
61	Asylum funding	3 (1%)	7
62	Mismanagement in public funds	3 (1%)	4
63	Clinical decision-making	3 (1%)	5
64	Crisis intervention	3 (1%)	5
65	Veterans' health	3 (1%)	5
66	Community care	3 (1%)	5
67	Medical research	3 (1%)	6
68	Bacon rationing	2 (1%)	4
69	Statutory fees	2 (1%)	2
70	Building cost reduction	2 (1%)	3
71	Distribution of health information through clinics and online	2 (1%)	4
72	role of medical officers of health	2 (1%)	3
73	Criminal justice - child homicide	2 (1%)	2
74	population displacement	2 (1%)	2
75	Professional misconduct investigations	2 (1%)	2
76	Detention and legal processes	2 (1%)	2
77	Blood product safety	2 (1%)	9
78	Friendly societies and financial protection	2 (1%)	3

#	Merged topic	Speeches	Phr.
79	Housing infrastructure maintenance	2 (1%)	3
80	GP access	2 (1%)	2
81	Union insolvency	2 (1%)	2
82	Housing equipment and supplies	2 (1%)	3
83	Judicial wellbeing	2 (1%)	6
84	Impact of education funding reforms	2 (1%)	4
85	Animal disease control	2 (1%)	4
86	at-risk groups identification	2 (1%)	4
87	Alcohol abuse impact	2 (1%)	16
88	Working class hardship	2 (1%)	3
89	Postgraduate medical training	2 (1%)	6
90	Extension of existing care provisions	2 (1%)	2
91	Impact of Covid-19	2 (1%)	5
92	Legal rights of vulnerable individuals	2 (1%)	2
93	Resident engagement in local decision-making	2 (1%)	2
94	Sickness benefit	2 (1%)	3
95	Non-medical clinicians in care decisions	2 (1%)	3
96	Medical error reporting	2 (1%)	5
97	Local government finance	2 (1%)	2
98	HIV/AIDS care and treatment	2 (1%)	5
99	Stigma in mental health	2 (1%)	5
100	cholera and diarrhoea treatment	2 (1%)	3

E. Most common sub-topic phrases, verbatim, before grouping. These show the raw vocabulary under the same expert HC→SC / JSON arm. Counts are lower than the clustered table because models rarely reuse an exact string.

Sub-theme phrase (verbatim)	Speeches	Total emissions
National Health Service	11	15
Poor Law administration	9	11
Disease prevention	6	9
Social welfare provision	6	6
NHS funding	5	9
Healthcare funding	5	6
Public health	5	5
Tuberculosis treatment	4	7
National Health Service funding	4	7
Mental health services	4	7
Workhouse conditions	4	6
Housing conditions	4	4
Mental Health	4	4
Public Health	4	4
Tuberculosis prevention	3	5
Carer support	3	5
Child welfare	3	4
public health	3	4
poor relief	3	3
Poor Law reform	3	3

F. Pilot facts. 270 speeches, read by four models under eight prompt wordings each (32 reads per speech at temperature zero), producing 8,692 distinct sub-topic phrases and 15,634 total phrase emissions on that core grid. Controlled experiments matched to it on speech, model and format accompany it: the uncapped arm behind Section 5’s cap finding (2,160 reads), the era-neutral definition arm (2,160 reads), and the expert-definition arm in both sentence orders (4,320 reads). Every positive read also returned free-text sub-topics; the topic map in Section 4 is built from the expert HC→SC definition under structured output (1,412 distinct phrases). The sample spans

1803 to the present. Sub-topics are generated by the models rather than chosen from a list. All figures come from the current pilot run.